

Notes on quantum partial search algorithm

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Abstract

The project aims to explore the efficiency of quantum partial search algorithm.

1 Grover's algorithm

Suppose that there are $N = 2^n$ items in the database, which can be represented by n qubits. Suppose that there is only one solution denoted as t , satisfying

$$f(x) = \begin{cases} 1 & \text{if } x = t; \\ 0 & \text{if } x \neq t. \end{cases} \quad (1.1)$$

Grover's algorithm is realized by the oracle operator

$$O_t = \mathbb{1} - 2|t\rangle\langle t|, \quad (1.2)$$

and the diffusion operator

$$D_n = \mathbb{1} - 2|s_n\rangle\langle s_n|, \quad (1.3)$$

with the uniform distribution state $|s_n\rangle$ given by

$$|s_n\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} |j\rangle. \quad (1.4)$$

Here j is in the binary form $j \in \{0, 1\}^n$.

The initial state of Grover's algorithm is $|s_n\rangle$. The items $|j\rangle$ can be divided into two groups: target state $|t\rangle$ and the rest non-target states, where the latter can be normalized as

$$|t^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{j \neq t} |j\rangle. \quad (1.5)$$

Therefore, the initial state $|s_n\rangle$ in Eq. (1.4) can be rewritten as

$$|s_n\rangle = \sin \theta_1 |t\rangle + \cos \theta_1 |t^\perp\rangle. \quad (1.6)$$

with the angle

$$\sin \theta_1 = \frac{1}{\sqrt{N}}. \quad (1.7)$$

In the new basis $\{|t\rangle, |t^\perp\rangle\}$, the oracle O_t in Eq. (1.2) can be rewritten as

$$O_t = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1.8)$$

The diffusion operator D_n can be rewritten as

$$D_n = \begin{pmatrix} \cos 2\theta_1 & -\sin 2\theta_1 \\ -\sin 2\theta_1 & -\cos 2\theta_1 \end{pmatrix}. \quad (1.9)$$

Then we define the Grover operator as

$$G_n = -D_n O_t = \begin{pmatrix} \cos 2\theta_1 & \sin 2\theta_1 \\ -\sin 2\theta_1 & \cos 2\theta_1 \end{pmatrix}. \quad (1.10)$$

The power in k of Grover operator is

$$G_n^k = \begin{pmatrix} \cos(2k\theta_1) & \sin(2k\theta_1) \\ -\sin(2k\theta_1) & \cos(2k\theta_1) \end{pmatrix}. \quad (1.11)$$

Acting on the initial state $|s_n\rangle$ in Eq. (1.4) gives

$$G_n^k |s_n\rangle = \sin((2k+1)\theta_1)|t\rangle + \cos((2k+1)\theta_1)|t^\perp\rangle. \quad (1.12)$$

The probability of finding the target state $|t\rangle$ is

$$\Pr(k) = \sin^2((2k+1)\theta_1). \quad (1.13)$$

If we want to find the target state with high probability, then we want the amplitude of $|t\rangle$ close to 1. Therefore, we want the optimal k to be

$$(2k_{\text{opt}} + 1)\theta_1 \rightarrow \frac{\pi}{2}, \quad (1.14)$$

which gives

$$k_{\text{opt}} \rightarrow \frac{\pi}{4\theta_1} - \frac{1}{2}. \quad (1.15)$$

Since k_{opt} must be an integer number, we take the value

$$k_{\text{opt}} = \left\lfloor \frac{\pi}{4\theta_1} - \frac{1}{2} \right\rfloor. \quad (1.16)$$

Here $\lfloor a \rfloor$ denotes the closest integer of a . For example $\lfloor \pi \rfloor = 3$. If N is large, we can approximate θ as

$$\theta_1 = \arcsin\left(\frac{1}{\sqrt{N}}\right) \approx \frac{1}{\sqrt{N}}. \quad (1.17)$$

Therefore the optimal iteration number k_{opt} is

$$k_{\text{opt}} = \left\lfloor \frac{\pi\sqrt{N}}{4} - \frac{1}{2} \right\rfloor. \quad (1.18)$$

It suggests a quadratic speedup compared to the classical search algorithm.

2 Quantum partial search algorithm

2.1 Local diffusion operator

The diffusion operator D_n defined in Eq. (1.3) is acting on all the n qubits. We can renormalize the diffusion operator as

$$D_{n,m} = \mathbb{1}_{2^{n-m}} \otimes (\mathbb{1}_{2^m} - 2|s_m\rangle\langle s_m|). \quad (2.1)$$

So the operator $D_{n,m}$ is only acting on m qubits which is $m \leq n$. We call it the local diffusion operator. Correspondingly, we can define the Grover operator with the local diffusion operator, namely

$$G_{n,m} = -D_{n,m}O_t. \quad (2.2)$$

For simplicity, we omit the subscription n when there is no ambiguous

$$G_m \equiv G_{n,m}. \quad (2.3)$$

The local diffusion operator D_m naturally divides the target state $|t\rangle$ as

$$|t\rangle = |t_1\rangle \otimes |t_2\rangle, \quad (2.4)$$

where $|t_1\rangle$ is $n - m$ qubits and $|t_2\rangle$ is m qubits. Here $|t_1\rangle$ can be viewed as the block index of $|t\rangle$. The total number of blocks denoted as K is

$$K = 2^{n-m}. \quad (2.5)$$

The total number of items in each block is

$$b = 2^m. \quad (2.6)$$

Note that we have

$$N = bK. \quad (2.7)$$

2.2 $O(3)$ representation

Define the new basis

$$|ntt\rangle = \frac{1}{\sqrt{2^m - 1}} \sum_{j \neq t_2} |t_1\rangle \otimes |j\rangle, \quad (2.8)$$

$$|u\rangle = \frac{1}{\sqrt{2^n - 2^m}} \left(\sqrt{2^n} |s_n\rangle - |t\rangle - \sqrt{2^m - 1} |ntt\rangle \right). \quad (2.9)$$

We can see that the state $|ntt\rangle$ is the uniform distribution of states which have the string t_1 . The state $|u\rangle$ is the uniform distribution of the rest states. Similar as $|s_n\rangle$ in Eq. (1.6), we have

$$|t_1\rangle \otimes |s_m\rangle = \frac{1}{\sqrt{2^m}} |t_1\rangle \otimes |t_2\rangle + \sqrt{\frac{2^m - 1}{2^m}} |ntt\rangle. \quad (2.10)$$

Define the angle

$$\sin \theta_2 = \frac{1}{\sqrt{2^m}}, \quad (2.11)$$

Therefore, we have

$$|t_1\rangle \otimes |s_m\rangle = \sin \theta_2 |t\rangle + \cos \theta_2 |ntt\rangle. \quad (2.12)$$

In the new basis $\{|t\rangle, |ntt\rangle, |u\rangle\}$, the initial state $|s_n\rangle$ in Eq. (1.4) can be rewritten as

$$|s_n\rangle = \frac{1}{\sqrt{2^n}} |t\rangle + \sqrt{\frac{2^m - 1}{2^n}} |ntt\rangle + \sqrt{\frac{2^n - 2^m}{2^n}} |u\rangle. \quad (2.13)$$

Define the angle

$$\sin \gamma = \frac{1}{\sqrt{2^n - 2^m}}. \quad (2.14)$$

Therefore, we have

$$|s_n\rangle = \sin \gamma \sin \theta_2 |t\rangle + \sin \gamma \cos \theta_2 |ntt\rangle + \cos \gamma |u\rangle. \quad (2.15)$$

Moreover, in the new basis $\{|t\rangle, |ntt\rangle, |u\rangle\}$, the global diffusion operator is

$$D_n = - \begin{pmatrix} 2 \sin^2 \gamma \sin^2 \theta_2 - 1 & 2 \sin^2 \gamma \sin \theta_2 \cos \theta_2 & 2 \sin \gamma \cos \gamma \sin \theta_2 \\ 2 \sin^2 \gamma \sin \theta_2 \cos \theta_2 & 2 \sin^2 \gamma \cos^2 \theta_2 - 1 & 2 \sin \gamma \cos \gamma \cos \theta_2 \\ 2 \sin \gamma \cos \gamma \sin \theta_2 & 2 \sin \gamma \cos \gamma \cos \theta_2 & 2 \cos^2 \gamma - 1 \end{pmatrix}. \quad (2.16)$$

Therefore, the global Grover operator G_n in Eq. (1.10) has the form

$$G_n = \begin{pmatrix} 1 - 2 \sin^2 \gamma \sin^2 \theta_2 & 2 \sin^2 \gamma \sin \theta_2 \cos \theta_2 & 2 \sin \gamma \cos \gamma \sin \theta_2 \\ -2 \sin^2 \gamma \sin \theta_2 \cos \theta_2 & 2 \sin^2 \gamma \cos^2 \theta_2 - 1 & 2 \sin \gamma \cos \gamma \cos \theta_2 \\ -2 \sin \gamma \cos \gamma \sin \theta_2 & 2 \sin \gamma \cos \gamma \cos \theta_2 & 2 \cos^2 \gamma - 1 \end{pmatrix}. \quad (2.17)$$

The local Grover operator G_m in Eq. (2.2) performs the search in the space spanned by $\{|t\rangle, |ntt\rangle\}$. Therefore, it has the form

$$G_m = \begin{pmatrix} \cos 2\theta_2 & \sin 2\theta_2 & 0 \\ -\sin 2\theta_2 & \cos 2\theta_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (2.18)$$

Here G_n is element of $O(3)$; G_m is element of $SO(3)$.

The k iterations of G_m is

$$G_m^k = \begin{pmatrix} \cos(2k\theta_2) & \sin(2k\theta_2) & 0 \\ -\sin(2k\theta_2) & \cos(2k\theta_2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (2.19)$$

The k iteration of G_n is complicated. However, we can consider the large block limit $b \rightarrow \infty$ ($\theta_2 \rightarrow 0$). Then we have [5]

$$G_n^k = \begin{pmatrix} \cos(2k\theta_1) & \sin(2k\theta_1) \sin \gamma & \sin(2k\theta_1) \cos \gamma \\ -\sin(2k\theta_1) \sin \gamma & (-1)^k \cos^2 \gamma + \cos(2k\theta_1) \sin^2 \gamma & \sin \gamma \cos \gamma ((-1)^{k+1} + \cos(2k\theta_1)) \\ -\sin(2k\theta_1) \cos \gamma & \sin \gamma \cos \gamma ((-1)^{k+1} + \cos(2k\theta_1)) & (-1)^k \sin^2 \gamma + \cos(2k\theta_1) \cos^2 \gamma \end{pmatrix}. \quad (2.20)$$

We can check that the G_n^k with the large block limit is still a valid orthogonal matrix, namely

$$G_n^{kT} G_n^k = G_n^k G_n^{kT} = \mathbb{I}. \quad (2.21)$$

Therefore, its inverse is

$$G_n^{k-1} = G_n^{kT} = \begin{pmatrix} \cos(2k\theta_1) & -\sin(2k\theta_1) \sin \gamma & -\sin(2k\theta_1) \cos \gamma \\ \sin(2k\theta_1) \sin \gamma & (-1)^k \cos^2 \gamma + \cos(2k\theta_1) \sin^2 \gamma & \sin \gamma \cos \gamma ((-1)^{k+1} + \cos(2k\theta_1)) \\ \sin(2k\theta_1) \cos \gamma & \sin \gamma \cos \gamma ((-1)^{k+1} + \cos(2k\theta_1)) & (-1)^k \sin^2 \gamma + \cos(2k\theta_1) \cos^2 \gamma \end{pmatrix}. \quad (2.22)$$

2.3 GRK partial search algorithm

The quantum partial search algorithm aims to find the block index of the target state. Therefore, the amplitude of $|u\rangle$ must vanish. The Grover-Radhakrishnan-Korepin partial search use the operator $G_n G_m^{k_2} G_n^{k_1}$ [1, 2, 3]. The number of iterations k_1 and k_2 must satisfy

$$\langle u | G_n G_m^{k_2} G_n^{k_1} | s_n \rangle = 0. \quad (2.23)$$

Suppose the scaling

$$k_1 = \frac{\pi}{4} \sqrt{N} - \eta \sqrt{b}, \quad k_2 = \alpha \sqrt{b}. \quad (2.24)$$

In the large block limit, the constrain of Eq. (2.23) becomes [2]

$$\tan\left(\frac{2\eta}{\sqrt{K}}\right) = \frac{2\sqrt{K} \sin 2\alpha}{K - 4 \sin^2 \alpha}. \quad (2.25)$$

The total number of oracle operators is

$$k_{\text{tot}} = \frac{\pi}{4} \sqrt{N} - \eta \sqrt{b} + \alpha \sqrt{b} = \frac{\pi}{4} \sqrt{N} - (\eta - \alpha) \sqrt{b}. \quad (2.26)$$

We want to maximum $(\eta - \alpha)$ with the constrain (2.25). The optimal value η and α can be numerically solved. Note that different K gives different optimal η and α .

2.4 Other operators

There are no solid arguments that the GRK algorithm (using $G_n G_m^{k_2} G_n^{k_1}$ with the optimal k_1 and k_2) is optimal. In other words, other operators may find a partial solution with fewer oracle operators. In [4], the following operators have been studied.

- The local-global operator $G_n^{k_2} G_m^{k_1}$. It can find the partial solution faster than Grover's algorithm, but it is slower than GRK algorithm.
- The global-local-global operator $G_n^{k_3} G_m^{k_2} G_n^{k_1}$. The GRK operator $G_n G_m^{k_2} G_n^{k_1}$ is a special case of the global-local-global operator with $k_3 = 1$. The optimization gives the GRK algorithm.
- The global-local-global-local operator $G_n G_m^{k_3} G_n^{k_2} G_m^{k_1}$. The iteration number of the last global operator is set as 1. The optimization shows that $k = 1$ gives the fastest algorithm, which is the GRK algorithm.

Table 1: Success probabilities of quantum partial search algorithm without local diffusion operators.

m		2	3	4	5	6	7
$k_{\text{tot}} = 2$	Pr (%)	10.5289	11.9491	14.7894	20.4701	31.8315	54.5544
$k_{\text{tot}} = 3$	Pr (%)	18.9371	20.2238	22.7972	27.9441	38.2378	58.8252
$k_{\text{tot}} = 4$	Pr (%)	29.3158	30.4377	32.6817	37.1696	46.1453	64.0968
$k_{\text{tot}} = 5$	Pr (%)	41.0188	41.9550	43.8274	47.5722	55.0619	70.0413
$k_{\text{tot}} = 6$	Pr (%)	53.3175	54.0585	55.5405	58.5045	64.4324	76.2883
$k_{\text{tot}} = 7$	Pr (%)	65.4463	65.9948	67.0918	69.2856	73.6734	82.4489
$k_{\text{tot}} = 8$	Pr (%)	76.6502	77.0208	77.7620	79.2446	82.2097	88.1398
$k_{\text{tot}} = 9$	Pr (%)	86.2315	86.4501	86.8872	87.7613	89.5097	93.0065
$k_{\text{tot}} = 10$	Pr (%)	93.5938	93.6955	93.8989	94.3056	95.1191	96.7461
$k_{\text{tot}} = 11$	Pr (%)	98.2788	98.3061	98.3608	98.4701	98.6886	99.1257

2.5 Conjecture

The unsolved problem is whether the GRK algorithm is the optimal quantum partial search algorithm. We have the following conjecture [5]

Conjecture 1. *Start from an arbitrary vector $|\phi\rangle$ in the three-dimensional space with the basis $\{|t\rangle, |ntt\rangle, |u\rangle\}$. If the sequence of local-global-local operations can find the target block*

$$\langle u | G_n^{k_4} G_m^{k_3} G_n^{k_2} G_m^{k_1} | \phi \rangle = 0, \quad (2.27)$$

then there exists a global-local-global sequence such that

$$\langle u | G_n^{\tilde{k}_3} G_m^{\tilde{k}_2} G_n^{\tilde{k}_1} | \phi \rangle = 0, \quad (2.28)$$

and it finds the target block faster, i.e.,

$$\tilde{k}_3 + \tilde{k}_2 + \tilde{k}_1 \leq k_1 + k_2 + k_3 + k_4. \quad (2.29)$$

Suppose that we have a partial search algorithm, given by

$$\langle u | G_n^{k_n} G_m^{k_{n-1}} G_n^{k_{n-2}} \dots G_m^{k_2} G_n^{k_1} | s_n \rangle = 0. \quad (2.30)$$

If the conjecture holds, we can derive a search operator (by applying the conjecture many times), given by

$$\langle u | G_n^{\tilde{k}_3} G_m^{\tilde{k}_2} G_n^{\tilde{k}_1} | s_n \rangle = 0, \quad (2.31)$$

which finds the target block faster. From [4], we know that the GRK algorithm is the fastest in all the global-local-global operators. Then we prove that GRK is the optimal partial search algorithm. Note that the conjecture is only a sufficient condition to prove the optimum of GRK algorithm. If the conjecture is not true, it does not mean that the GRK algorithm is not optimal. But if the GRK is not optimal, then the conjecture is not valid.

3 Numerical quests

GRK algorithm is designed to find the target block with the success probability 1 (in the large database limit). However, it is unknown the highest success probability of finding the target block given the iteration number k . In other words, we aim to find the optimal search operator with k oracle operators.

When $n = 8$, namely $N = 2^8$, Grover's algorithm requires $k_{\text{opt}} = 12$ oracle operators. The success probability is around 99.9947%.

Table 2: Optimal operators and success probabilities of quantum partial search algorithm with $n = 8$.

m		2	3	4	5	6	7
$k_{\text{tot}} = 2$	Operator	$G_8 G_2$	$G_8 G_3$	$G_8 G_4$	$G_8 G_5$	$G_8 G_6$	$G_8 G_7$
	Pr (%)	10.5747	13.4310	16.9145	22.7749	33.9446	56.0089
$k_{\text{tot}} = 3$	Operator	$G_8 G_2 G_8$	$G_8 G_3 G_8$	$G_8 G_4^2$	$G_8 G_5^2$	$G_8 G_6^2$	$G_8 G_7^2$
	Pr (%)	19.0236	23.0100	27.0968	33.6418	43.8606	62.8228
$k_{\text{tot}} = 4$	Operator	$G_8 G_2 G_8^2$	$G_8 G_3 G_8^2$	$G_8 G_4^2 G_8$	$G_8 G_5^3$	$G_8 G_6^3$	$G_8 G_7^3$
	Pr (%)	29.4353	34.2693	38.7202	45.7859	55.6425	71.1470
$k_{\text{tot}} = 5$	Operator	$G_8 G_2 G_8^3$	$G_8 G_3 G_8^3$	$G_8 G_4^2 G_8^2$	$G_8 G_5^3 G_8$	$G_8 G_6^4$	$G_8 G_7^4$
	Pr (%)	41.1617	46.5081	51.0612	58.2046	67.7145	79.9491
$k_{\text{tot}} = 6$	Operator	$G_8 G_2 G_8^4$	$G_8 G_3 G_8^4$	$G_8 G_4^2 G_8^3$	$G_8 G_5^3 G_8^2$	$G_8 G_6^4 G_8$	$G_8 G_7^5$
	Pr (%)	53.4727	58.9644	63.3514	70.1245	78.7004	88.1374
$k_{\text{tot}} = 7$	Operator	$G_8 G_2 G_8^5$	$G_8 G_3 G_8^5$	$G_8 G_4^2 G_8^4$	$G_8 G_5^3 G_8^3$	$G_8 G_6^4 G_8^2$	$G_8 G_7^6$
	Pr (%)	65.6019	70.8626	74.8257	80.8038	87.9164	94.6963
$k_{\text{tot}} = 8$	Operator	$G_8 G_2 G_8^6$	$G_8 G_3 G_8^6$	$G_8 G_4^2 G_8^5$	$G_8 G_5^3 G_8^4$	$G_8 G_6^4 G_8^3$	$G_8 G_7^7$
	Pr (%)	76.7941	81.4622	84.7698	89.5775	94.7887	98.8124
$k_{\text{tot}} = 9$	Operator	$G_8 G_2 G_8^7$	$G_8 G_3 G_8^7$	$G_8 G_4^2 G_8^6$	$G_8 G_5^3 G_8^5$	$G_8 G_6^4 G_8^4$	$G_8 G_7^6 G_8 G_7$
	Pr (%)	86.3525	90.1031	92.5645	95.8994	98.8894	99.9998
$k_{\text{tot}} = 10$	Operator	$G_8 G_2 G_8^8$	$G_8 G_3 G_8^8$	$G_8 G_4^2 G_8^7$	$G_8 G_5^3 G_8^6$	$G_8 G_6^4 G_8^3 G_8^2$	$G_8 G_7^3 (G_8 G_7)^2$
	Pr (%)	93.6822	96.2474	97.7247	99.3760	99.9999	99.9999
$k_{\text{tot}} = 11$	Operator	$G_8 G_2 G_8^9$	$G_8 G_3 G_8^9$	$G_8 G_4^2 G_8^8$	$G_8 G_5 (G_8^3 G_5)^2 G_8 G_5^2$	$G_8 G_6 G_8^4 (G_6 G_8)^2 G_8$	$G_7 G_8 G_7^3 (G_8 G_7)^3$
	Pr (%)	98.3268	99.5126	99.9290	99.9999	99.9999	99.9999

4 Bounds on the success probability

4.1 Partial search without local diffusion operator

Theorem 1. Given k iterations of the global Grover's operator, the success probability of finding the target block t_1 is

$$\Pr(G_n^k, x = t_1) = \sin^2((2k+1)\theta_1) + \frac{b-1}{N-1} \cos^2((2k+1)\theta_1). \quad (4.1)$$

Proof. The normalized non-target state $|t^\perp\rangle$ can be rewritten as

$$|t^\perp\rangle = \frac{\sin \gamma \cos \theta_2}{\cos \theta_1} |ntt\rangle + \frac{\cos \gamma}{\cos \theta_1} |u\rangle. \quad (4.2)$$

Therefore, Eq. (4.12) can be rewritten as

$$G_n^k |s_n\rangle = \sin((2k+1)\theta_1) |t\rangle + \cos((2k+1)\theta_1) \frac{\sin \gamma \cos \theta_2}{\cos \theta_1} |ntt\rangle + \cos((2k+1)\theta_1) \frac{\cos \gamma}{\cos \theta_1} |u\rangle. \quad (4.3)$$

Then the success probability of finding t_1 is given by

$$\Pr(G_n^k, x = t_1) = 1 - |\langle u | G_n^k | s_n \rangle|^2 = 1 - \cos^2((2k+1)\theta_1) \frac{\cos^2 \gamma}{\cos^2 \theta_1}. \quad (4.4)$$

Note that

$$\cos \gamma = \sqrt{\frac{K-1}{K}}, \quad \cos \theta_1 = \sqrt{\frac{N-1}{N}}. \quad (4.5)$$

Therefore, the success probability can be rewritten as

$$\begin{aligned} \Pr(G_n^k, x = t_1) &= 1 - \cos^2((2k+1)\theta_1) \frac{K-1}{K} \frac{N}{N-1} \\ &= 1 - \frac{N-b}{N-1} \cos^2((2k+1)\theta_1) \\ &= 1 - \left(1 - \frac{b-1}{N-1}\right) \cos^2((2k+1)\theta_1) \\ &= \sin^2((2k+1)\theta_1) + \frac{b-1}{N-1} \cos^2((2k+1)\theta_1). \end{aligned} \quad (4.6)$$

Note that $N = bK$. □

Corollary 1.1. *Given k iterations of the global Grover's operator, the success probability of finding the target block is higher than finding the full solution. The success probability difference is*

$$\Delta \Pr = \Pr(G_n^k, x = t_1) - \Pr(G_n^k, x = t) = \frac{b-1}{N-1} \cos^2((2k+1)\theta_1), \quad (4.7)$$

which scales $\mathcal{O}(b/N)$.

4.2 Partial search with local diffusion operator

The numerical tests show that the operator $G_1 G_2^{k_2} G_1^{k_1}$ always gives the highest success probabilities given a fixed number of total oracle operators. See Table. 2.

Consider the large block limit, namely $b \rightarrow \infty$ and $\theta_2 \rightarrow 0$. The initial state is

$$|s_n\rangle = (0, \sin \gamma, \cos \gamma)^T + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right), \quad (4.8)$$

in the basis $\{|n\rangle, |ntt\rangle, |u\rangle\}$. First, acting $G_1^{k_1}$ on the initial state gives

$$G_1^{k_1} |s_n\rangle = \begin{pmatrix} \sin(2k_1\theta_1) \\ \sin \gamma \sin(2k_1\theta_1) \\ \cos \gamma \cos(2k_1\theta_1) \end{pmatrix} + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \quad (4.9)$$

Next, the $G_2^{k_2}$ operator gives

$$G_2^{k_2} G_1^{k_1} |s_n\rangle = \begin{pmatrix} \sin(2k_1\theta_1) \cos(2k_2\theta_2) + \sin \gamma \cos(2k_1\theta_1) \sin(2k_2\theta_2) \\ -\sin(2k_1\theta_1) \sin(2k_2\theta_2) + \sin \gamma \cos(2k_1\theta_1) \cos(2k_2\theta_2) \\ \cos \gamma \cos(2k_1\theta_1) \end{pmatrix} + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \quad (4.10)$$

In the large block limit, the global diffusion operator has the form

$$D_n = 2|s_n\rangle\langle s_n| - \mathbb{1} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\cos(2\gamma) & \sin(2\gamma) \\ 0 & \sin(2\gamma) & \cos(2\gamma) \end{pmatrix} + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \quad (4.11)$$

We can omit the oracle operator in the last iteration since we only consider the leading order of $1/\sqrt{b}$ or $1/\sqrt{N}$. Finally, we have

$$\begin{aligned} &\langle u | D_n G_2^{k_2} G_1^{k_1} | s_n \rangle \\ &= -\sin(2\gamma) \sin(2k_1\theta_1) \sin(2k_2\theta_2) + \sin(2\gamma) \sin \gamma \cos(2k_1\theta_1) \cos(2k_2\theta_2) + \cos \gamma \cos(2\gamma) \cos(2k_1\theta_1) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \end{aligned} \quad (4.12)$$

4.2.1 Large block number limit

We consider the large block number limit, namely $K \rightarrow \infty$ or $\gamma \rightarrow 0$. In the first order of γ , we have

$$\langle u | D_n G_2^{k_2} G_1^{k_1} | s_n \rangle = -2\gamma \sin(2k_1\theta_1) \sin(2k_2\theta_2) + \cos(2k_1\theta_1) + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \quad (4.13)$$

Suppose that the total number of iterations is fixed, given by

$$(k_1 + k_2)\theta_1 = \alpha. \quad (4.14)$$

Here α is a fixed number. Furthermore, suppose that

$$k_2\theta_2 = \beta, \quad (4.15)$$

which gives

$$k_1\theta_1 = \alpha - k_2\theta_1 = \alpha - \frac{\beta\theta_1}{\theta_2} = \alpha - \frac{\beta}{\sqrt{K}} = \alpha - \beta\gamma. \quad (4.16)$$

In the first order of γ , we have

$$\begin{aligned} \langle u | D_n G_2^{k_2} G_1^{k_1} | s_n \rangle &= -2\gamma \sin(2\alpha - 2\beta\gamma) \sin(2\beta) + \cos(2\alpha - 2\beta\gamma) + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right) \\ &= -2\gamma \sin(2\alpha) \sin(2\beta) + \cos(2\alpha) + 2\beta\gamma \sin(2\alpha) + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right) \\ &= \cos(2\alpha) + 2\sin(2\alpha)\gamma(\beta - \sin(2\beta)) + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \end{aligned} \quad (4.17)$$

Define the function

$$f(\beta) = \beta - \sin(2\beta), \quad (4.18)$$

which takes the minimum at $\beta = \pi/6$. And the minimal value is

$$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{\sqrt{3}}{2} \approx -0.342427. \quad (4.19)$$

Denote the minimal value as

$$\varepsilon = -2f\left(\frac{\pi}{6}\right). \quad (4.20)$$

Then we know that the success probability of finding the target block has the bound

$$\begin{aligned} \Pr(D_n G_2^{k_2} G_1^{k_1}, x = t_1) &= 1 - \left| \langle u | D_n G_2^{k_2} G_1^{k_1} | s_n \rangle \right|^2 \\ &\leq 1 - |(\cos(2\alpha) - \sin(2\alpha)\gamma\varepsilon)|^2 + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right) \\ &= \sin^2(2\alpha) + \sin(4\alpha)\varepsilon\gamma + \mathcal{O}(\gamma^2) + \mathcal{O}\left(\frac{1}{\sqrt{b}}\right). \end{aligned} \quad (4.21)$$

In conclusion, we have the following theorem

Theorem 2. Given $k = \alpha\sqrt{N}$ oracle operators, the success probability of finding the target block has the upper bound

$$\Pr(x = t_1) \leq \sin^2(2\alpha) + \sin(4\alpha)\varepsilon\gamma + \mathcal{O}(\gamma^2), \quad (4.22)$$

with the constant $\varepsilon = \sqrt{3} - \pi/3 \approx 0.68485$.

Corollary 2.1. Given $k = \alpha\sqrt{N}$ oracle operators, the success probability difference between finding the full solution and the target block has the upper bound

$$\Pr(x = t_1) - \Pr(x = t) \leq \sin(4\alpha)\varepsilon\gamma + \mathcal{O}(\gamma^2), \quad (4.23)$$

with the constant $\varepsilon = \sqrt{3} - \pi/3 \approx 0.68485$. The probability difference scales $\mathcal{O}(1/\sqrt{K})$.

5 Bounds on the minimal expected number of oracles

5.1 Partial search without local diffusion operator

Given $k + 1/2 = \alpha/\theta_1$ iterations of Grover operator, the success probability of finding the full solution is given by Eq. (1.13), namely

$$\Pr(x = t) = \sin^2(2\alpha). \quad (5.1)$$

The expected number of oracles is

$$E(G_1^k) = \frac{\alpha}{\theta_1 \sin^2(2\alpha)} + \mathcal{O}\left(\frac{1}{N}\right). \quad (5.2)$$

The minimal point in the leading order is given by

$$\frac{dE(G_1^k)}{d\alpha} = \frac{\sin(2\alpha) - 4\alpha \cos(2\alpha)}{\theta_1 \sin^3(2\alpha)} = 0, \quad (5.3)$$

which gives

$$4\alpha_{\min} = \tan(2\alpha_{\min}). \quad (5.4)$$

The approximate numerical solution is

$$\alpha_{\min} \approx 0.582781, \quad \sin^2(2\alpha_{\min}) \approx 0.844579, \quad \frac{\alpha_{\min}}{\sin^2(2\alpha_{\min})} \approx 0.690025. \quad (5.5)$$

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